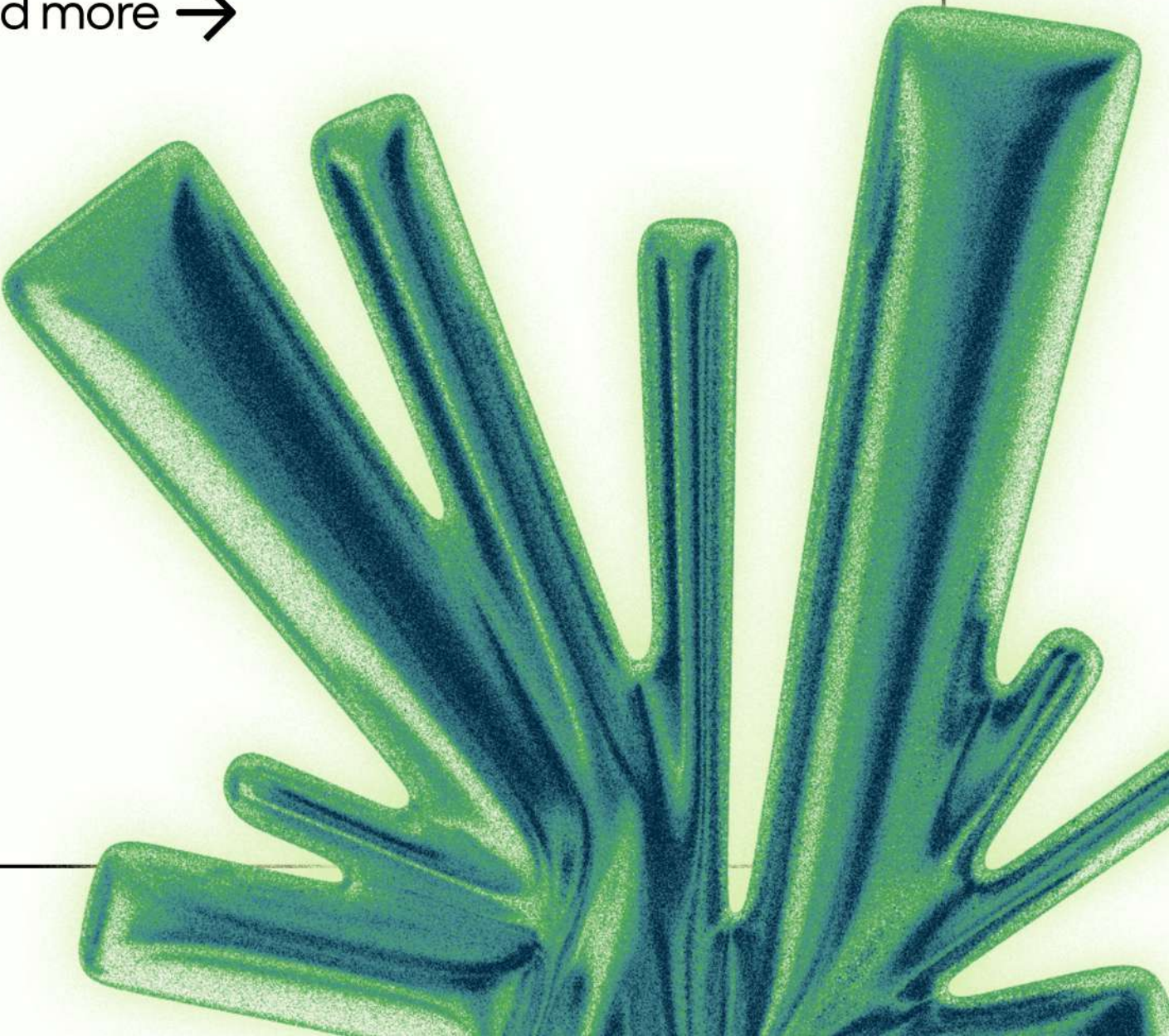




AI Model Architectures Explained

Swipe to read more →





Large Language Models

A super-autocomplete engine that reasons using attention.

How it processes information:

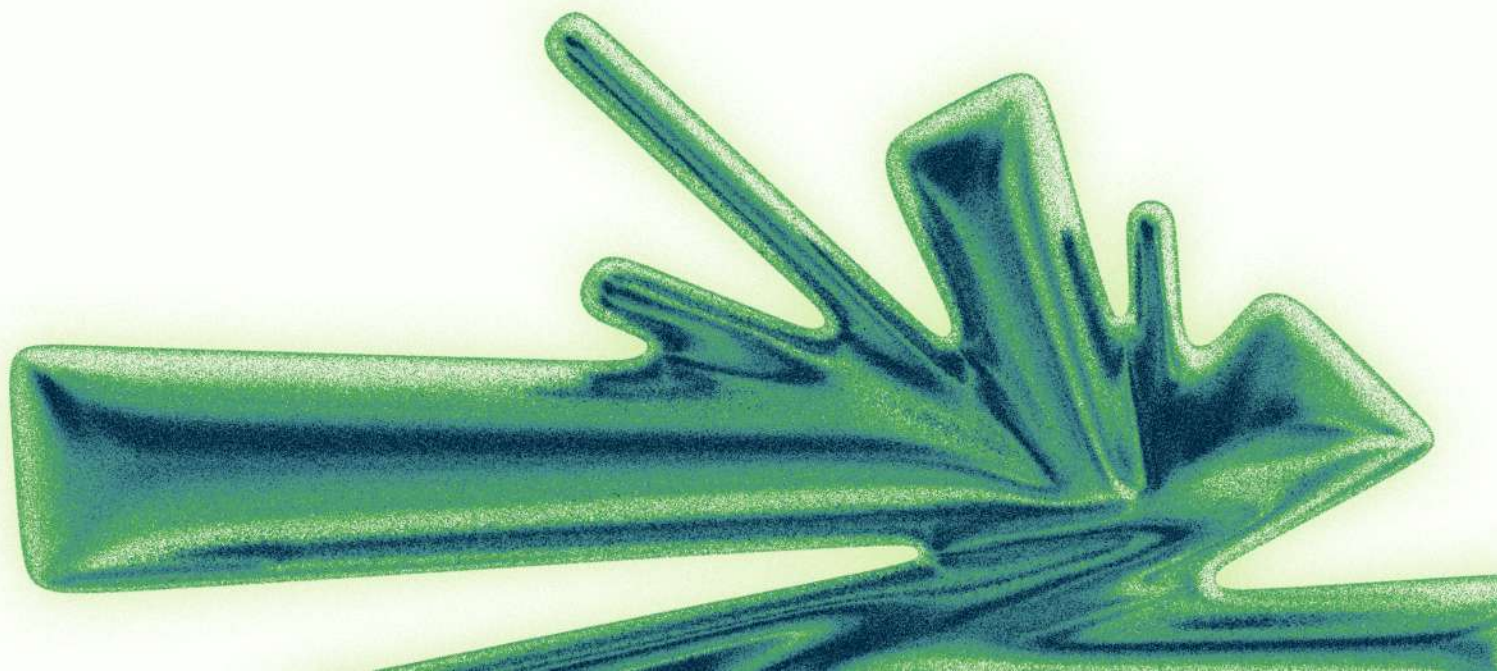
- Breaks text into tiny pieces (tokens)
- Looks at all relevant tokens at once (self-attention = "focus where needed")
- Builds meaning layer by layer
- Predicts the next token

Why it works:

LLMs don't understand words, they understand relationships between pieces of text.

Think of it as:

A very smart "next-word predictor" that learned logic by reading the entire internet.





Large Concept Models

Thinks in full ideas, not tokens – like jumping from paragraph to concept.

How it processes information:

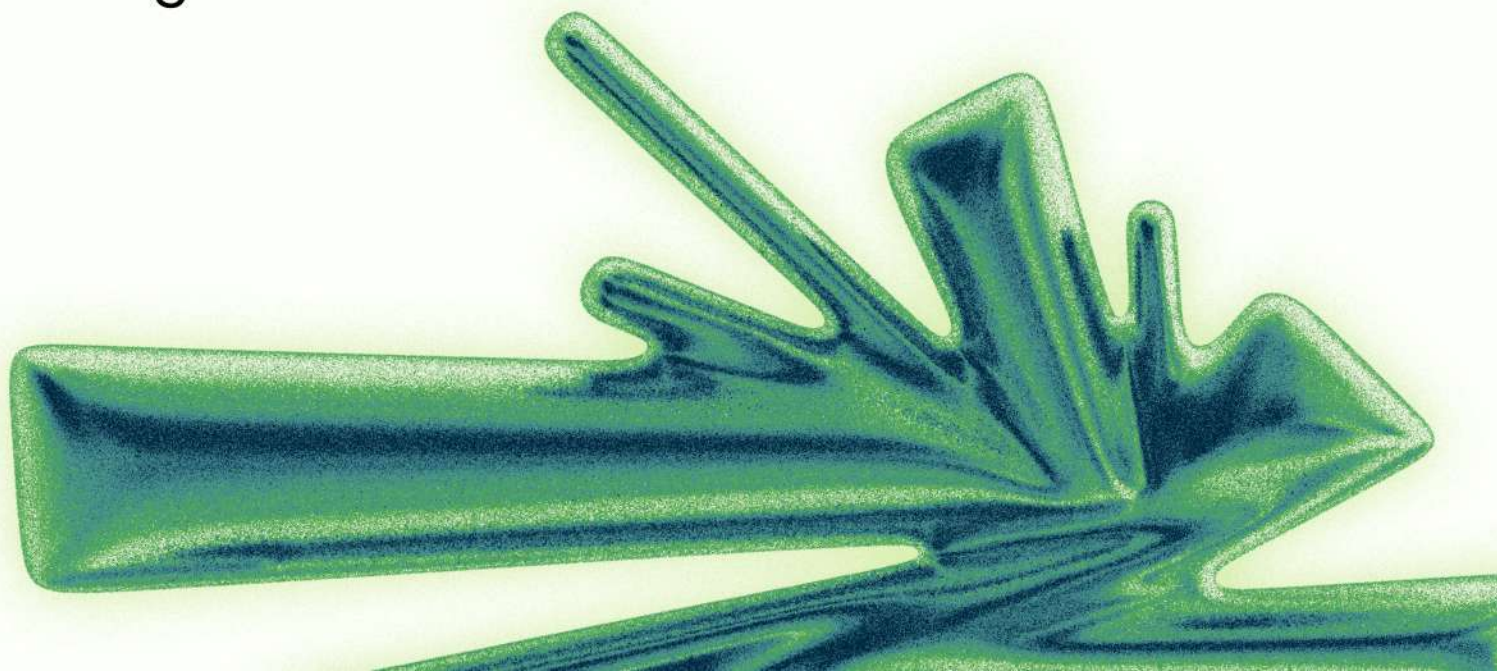
- Splits text into meaningful chunks (sentences, paragraphs)
- Turns them into concept embeddings (SONAR)
- Uses diffusion to refine the meaning (gradually denoise the idea)

Why it works:

LCMS skip token-by-token noise and operate directly on meaning.

Think of it as:

An LLM that doesn't think letter word → sentence, but concept → reasoning → answer.





Large Action Models

An AI that doesn't just reply – it actually does things.

How it processes information:

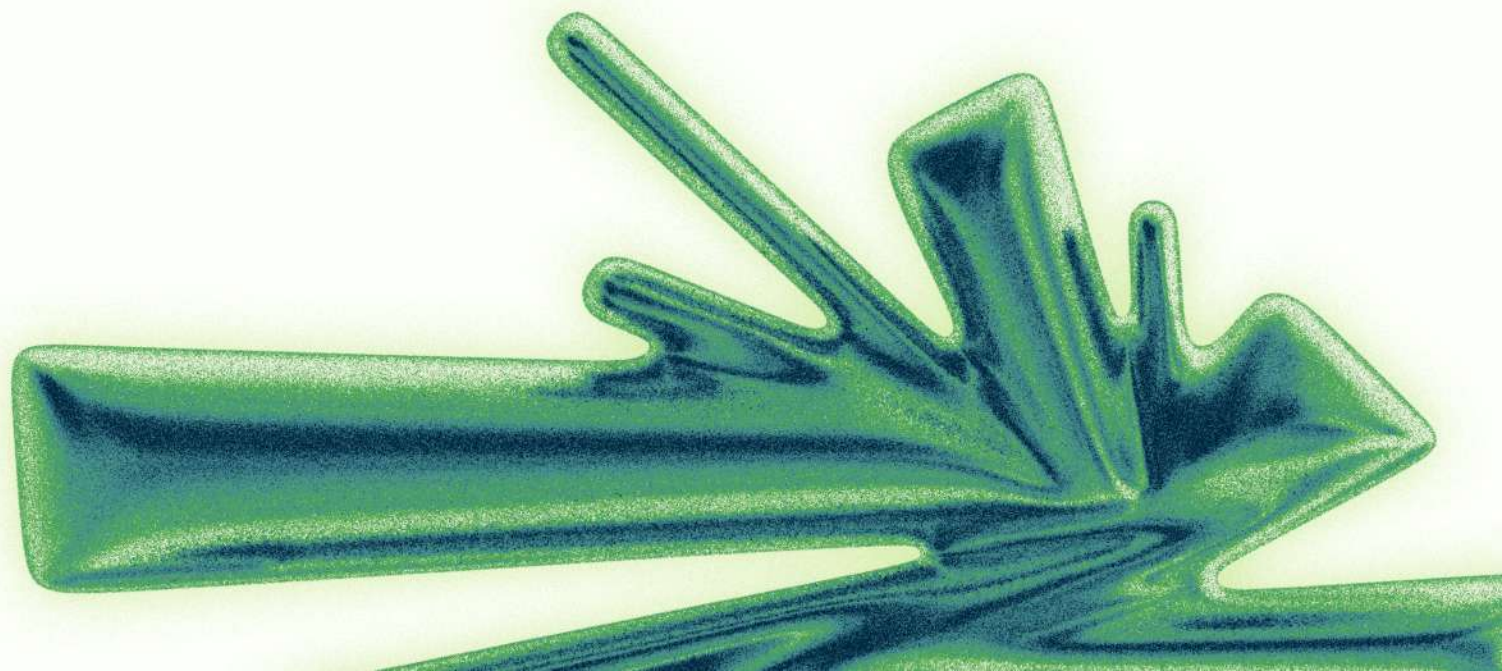
- Sees the screen/context
- Understands what you want (intent)
- Breaks it into steps
- Plans & executes actions
- Checks if the action worked
- Tries again if needed

Why it works:

LAMs include perception + planning + memory – the three elements needed for agency.

Think of it as:

A digital assistant that can operate your computer like a human.





Mixture of Experts

A classroom full of experts – the router picks who answers each question.

How it processes information:

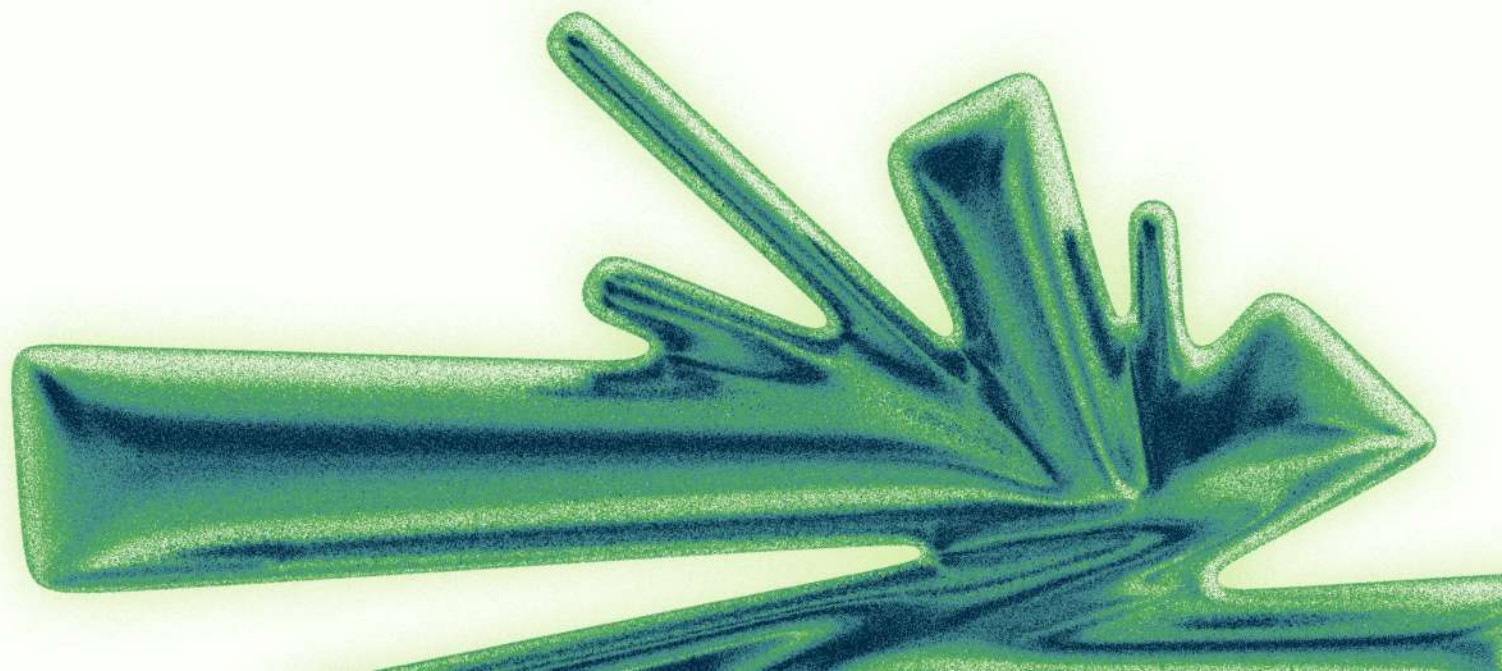
- Router looks at your input
- Activates only the experts best suited for it
- Each expert gives its answer
- Model combines their outputs

Why it works:

You get the power of a huge model without running the whole thing every time.

Think of it as:

Instead of one giant brain, it's a team of specialists.





Vision-Language Models

The model sees (vision encoder) and understands (text encoder), then combines both.

How it processes information:

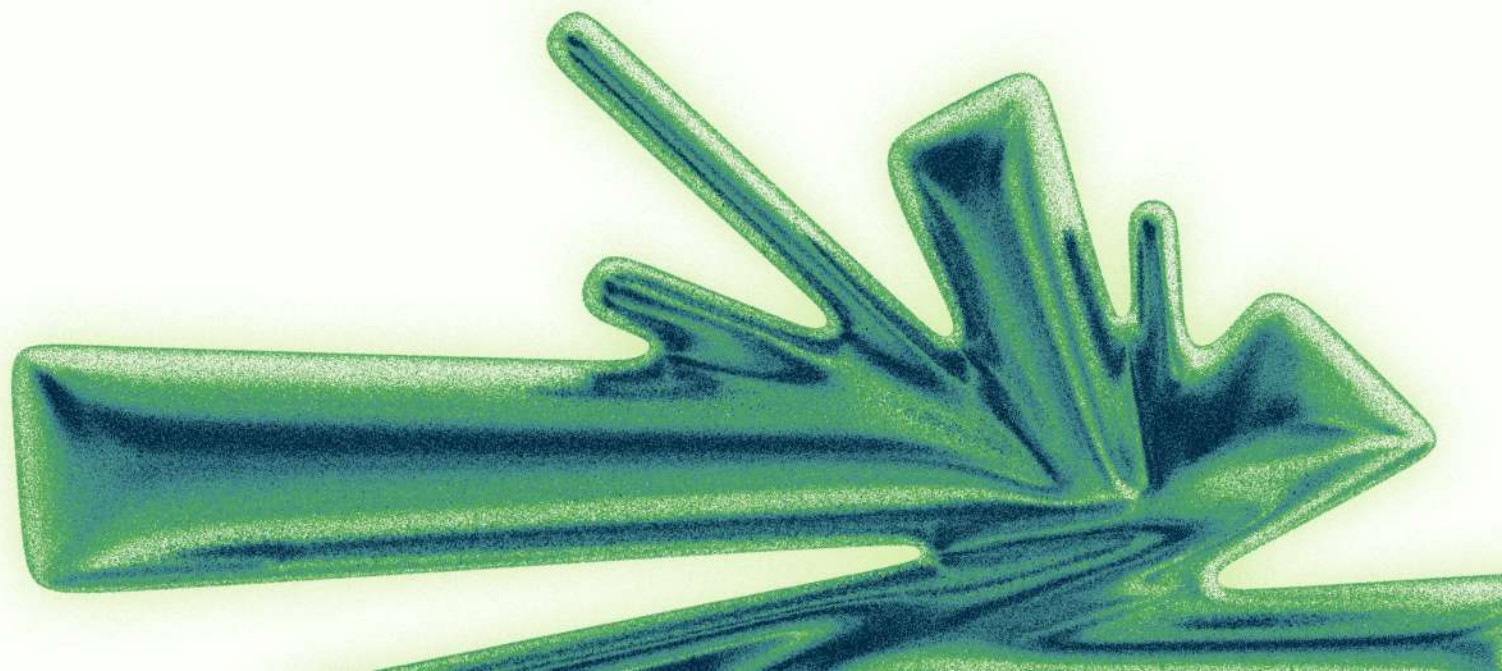
- Image → visual patches
- Text → token embeddings
- Fusion layer aligns both spaces
- LLM generates the answer

Why it works:

It turns pixels into meaning, then meaning into language.

Think of it as:

The AI version of "look at this and explain it to me."





Small Language Models

Compact, efficient models optimized for speed and specific tasks.

How it processes information:

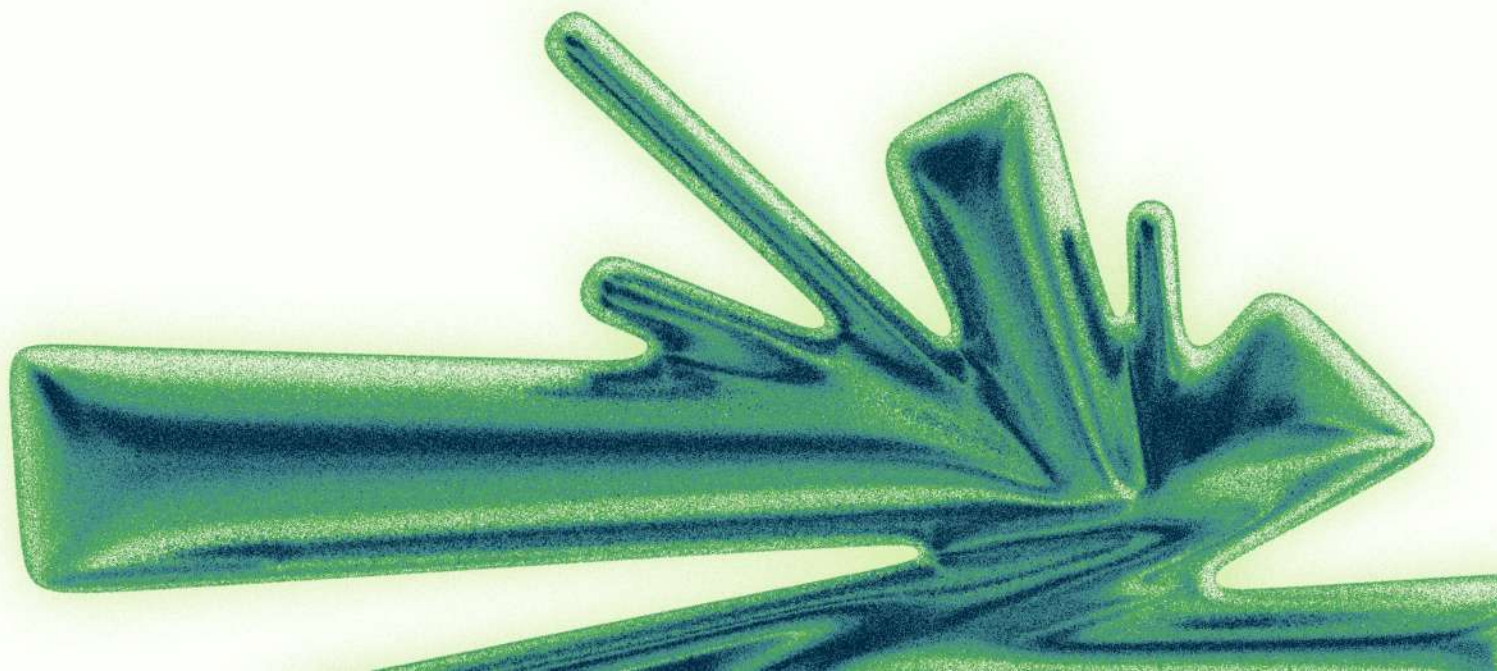
- Same transformer architecture as LLMs
- Fewer parameters (typically under 10B)
- Often distilled from larger models
- Fine-tuned for narrow use cases

Why it works:

SLMs trade broad knowledge for efficiency, making them fast enough to run on-device or at scale.

Think of it as:

A specialist that knows one thing really well instead of knowing a little about everything.





Masked Language Models

Reads sentences like a human: sees the whole thing, predicts hidden words.

How it processes information:

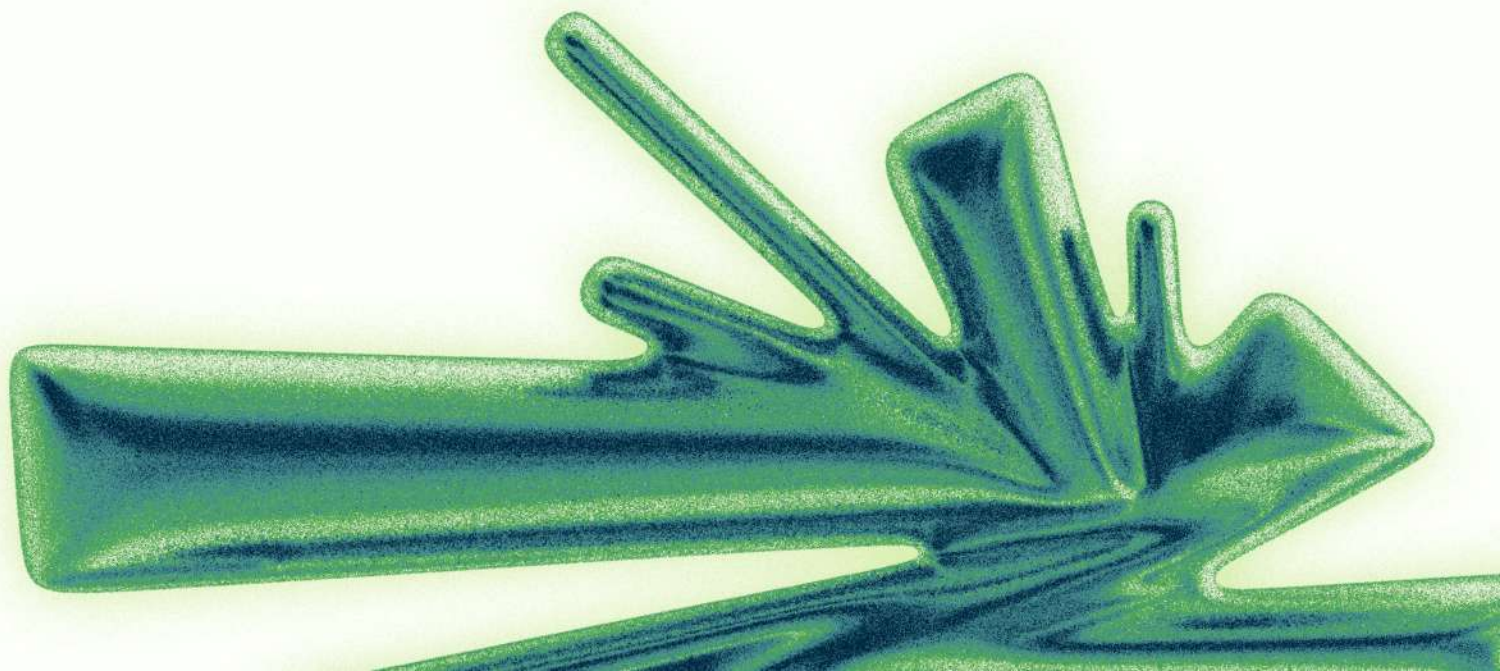
- Masks some tokens
- Looks at full left + right context
- Predicts missing text
- Builds contextual embeddings

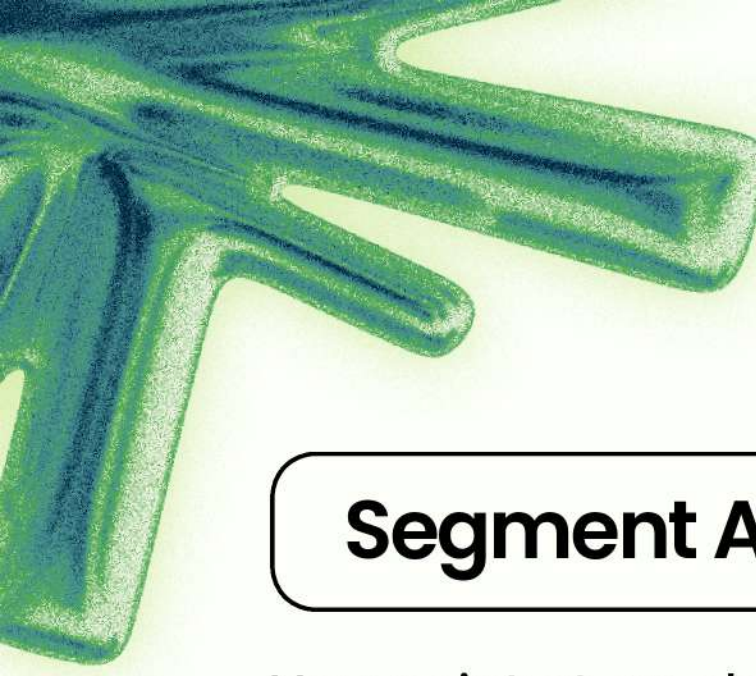
Why it works:

Its bidirectional view gives it deep semantic understanding.

Think of it as:

A language model trained on "fill in the blanks."





Segment Anything Models

You point at an object → SAM outlines it perfectly, instantly.

How it processes information:

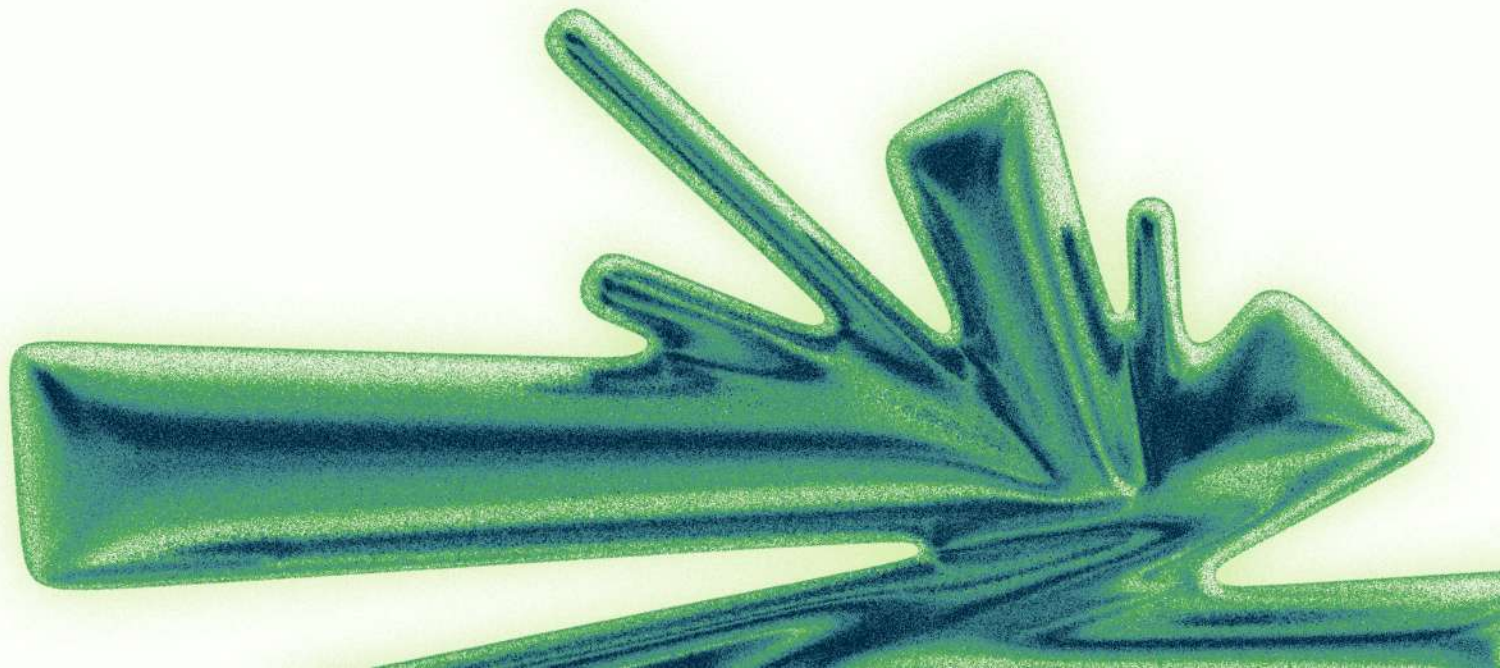
- Vision Transformer encodes the entire image
- Prompt encoder encodes your point/box
- Mask decoder merges both
- Output = pixel-accurate segmentation

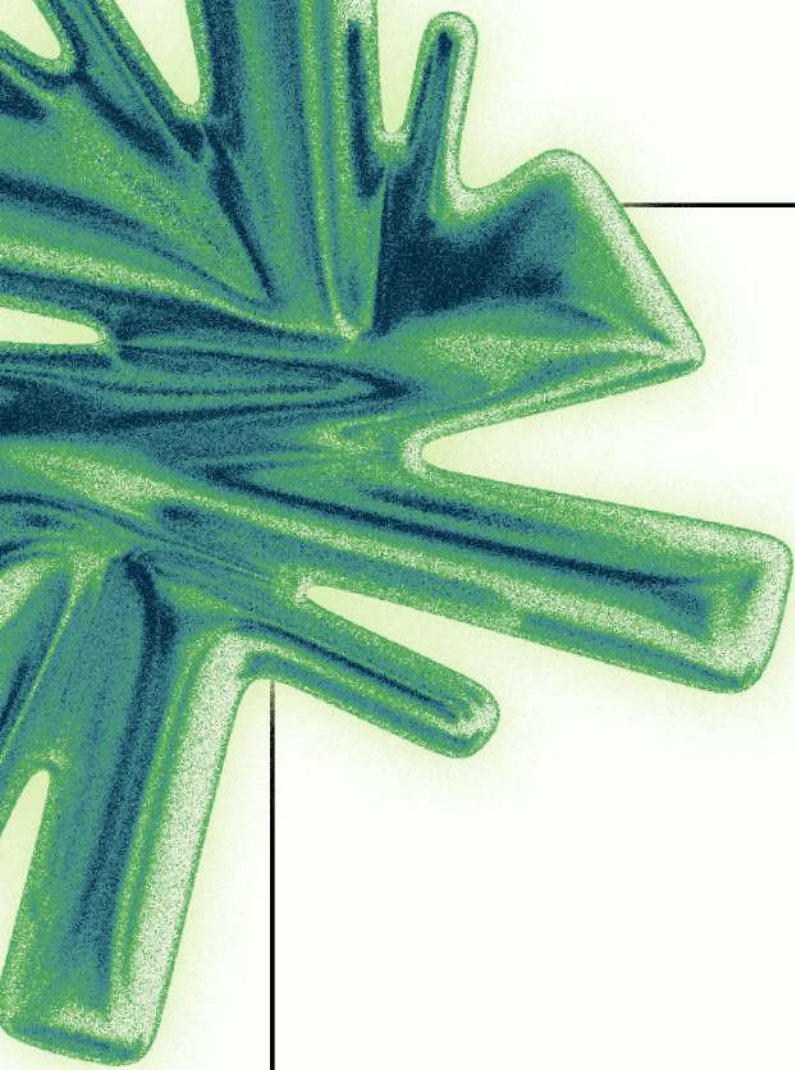
Why it works:

SAM generalizes across images – no retraining needed.

Think of it as:

A universal "select object" tool for the real world.





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